Research Progress Report Presentation, August, 2026

GreenMorph: Sustainable Neuromorphic Computing through Energy-Harvesting and Energy-Driven Online STDP　Learning

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Outline

- Introduction
- Proposed Framework
- Evaluation
- Conclusion
- Schedule

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Introduction

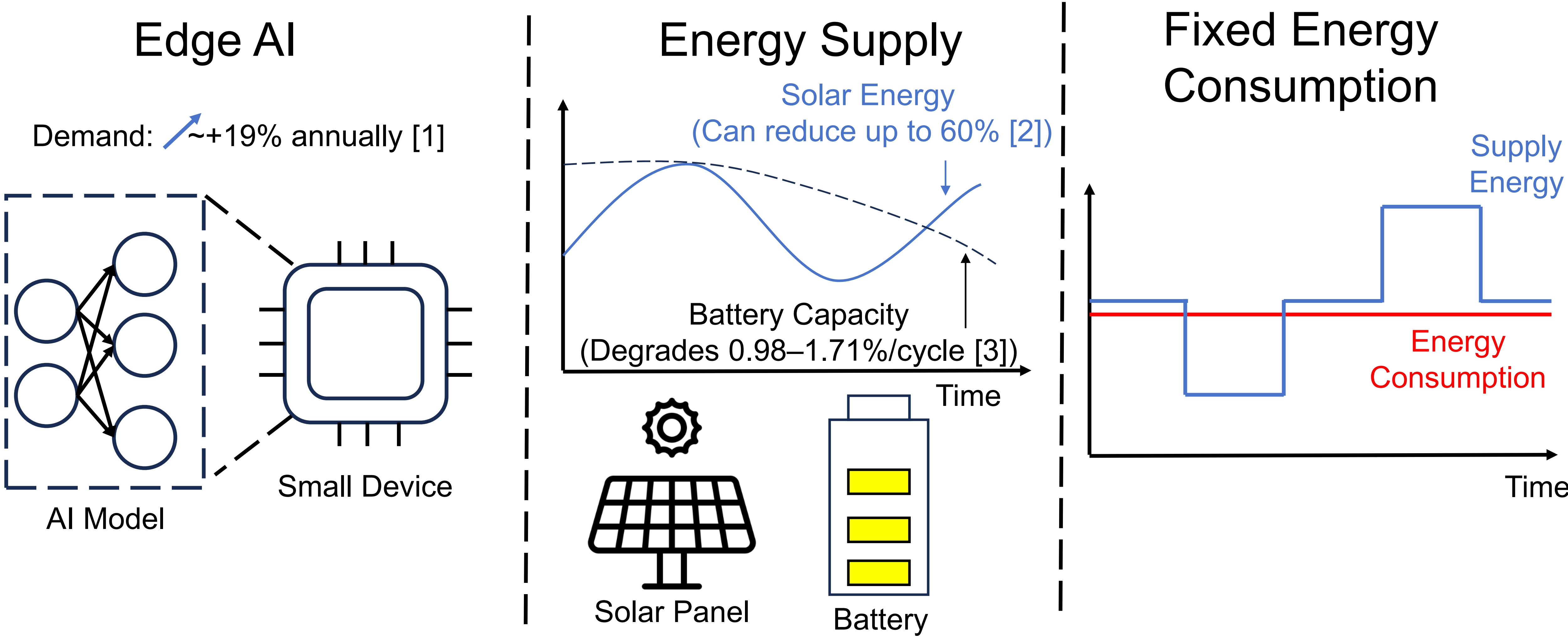
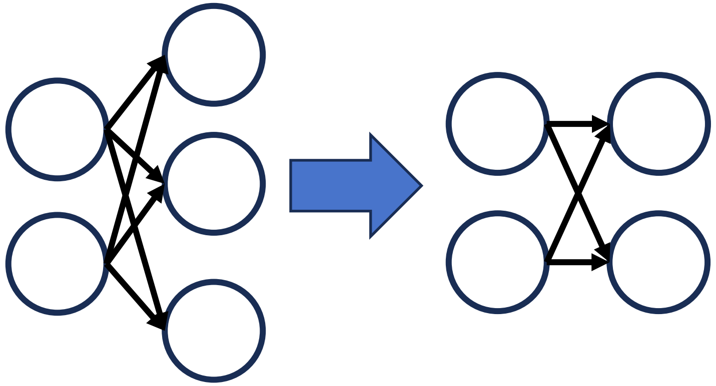


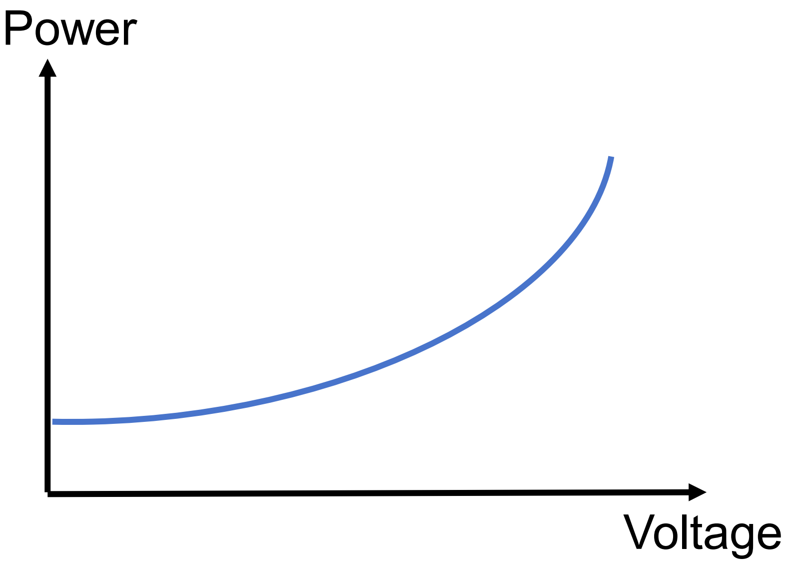
Fig. 1 Edge AI and Energy

Introduction (cnt.)

Model Downscaling [4]



Voltage/Frequency Scaling [5]



Weight Quantization [6]

Floating Point

1.2	2.3
1.5	3.1

Integer

1	2
2	3

Fig. 2 Common Energy Saving Methods

Introduction (cnt.)

Spiking Neural Networks (SNNs) [7]

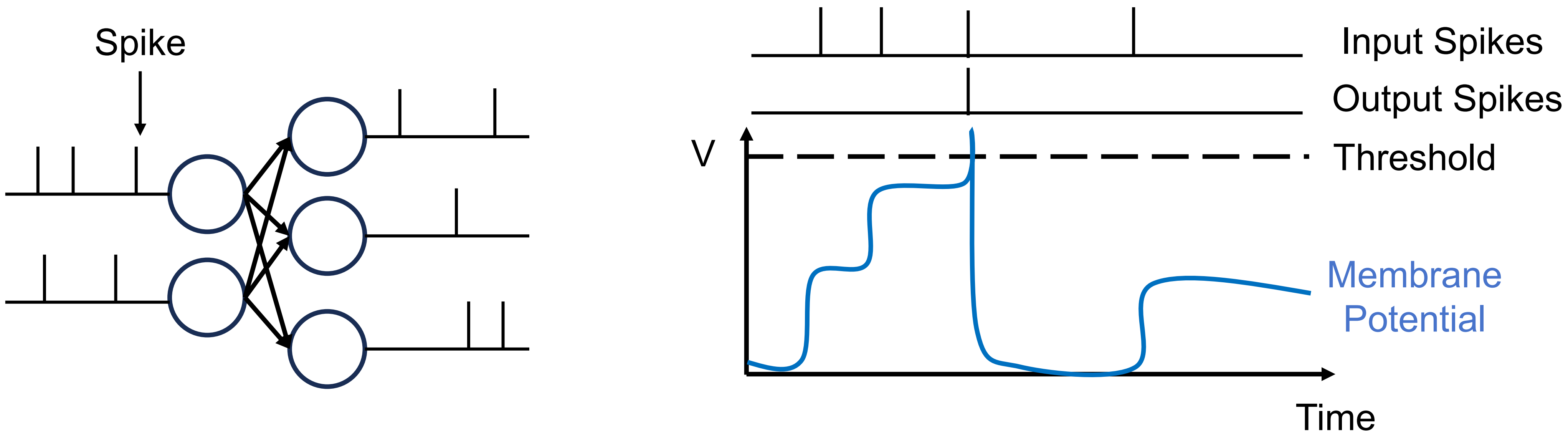
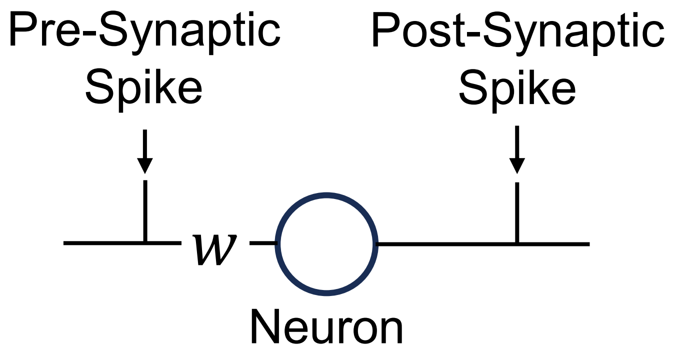


Fig. 3 Spiking Neural Networks

Introduction (cnt.)

Spike-Timing-Dependent-Plasticity (STDP) [8]



$\mathrm{t_{post}} > \mathrm{t_{pre}} \Rightarrow \Delta w \nearrow$

$\mathrm{t_{post}} = \mathrm{t_{pre}} \Rightarrow \Delta w$ No Update

$\mathrm{t_{post}} < \mathrm{t_{pre}} \Rightarrow \Delta w \searrow$

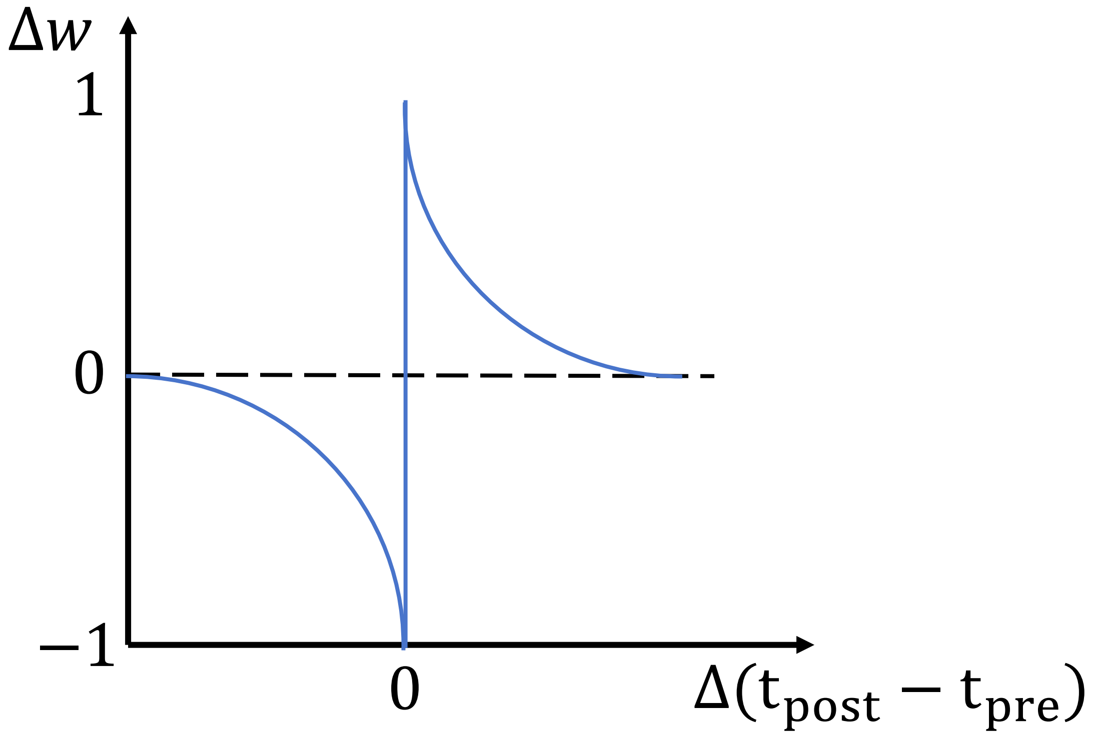


Fig. 4 Spike-Timing-Dependent-Plasticity

Introduction (cnt.)

Challenges:

Unstable energy supply for edge AI systems.

Simple energy reduction leads to suboptimal performance.

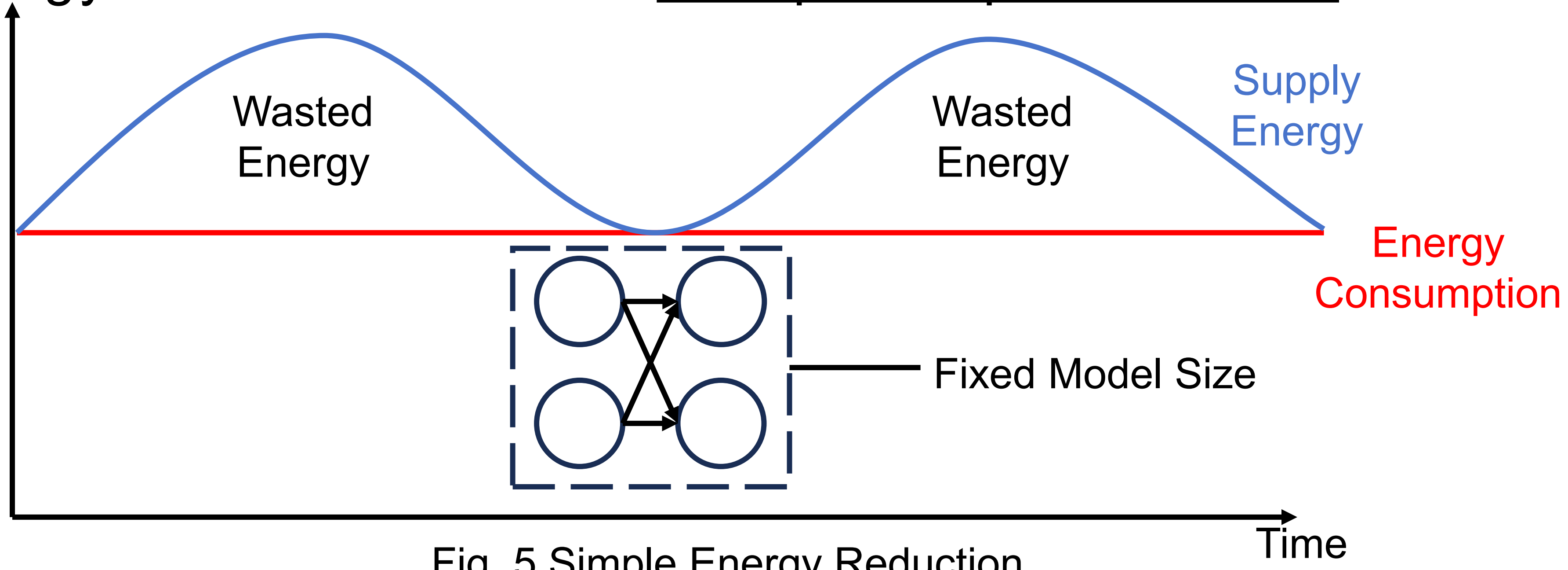


Fig. 5 Simple Energy Reduction

Introduction (cnt.)

Our contributions:

Dynamic energy adaptation framework for Spiking Neural Networks.

Similarity-based neuron pruning to minimize performance loss.

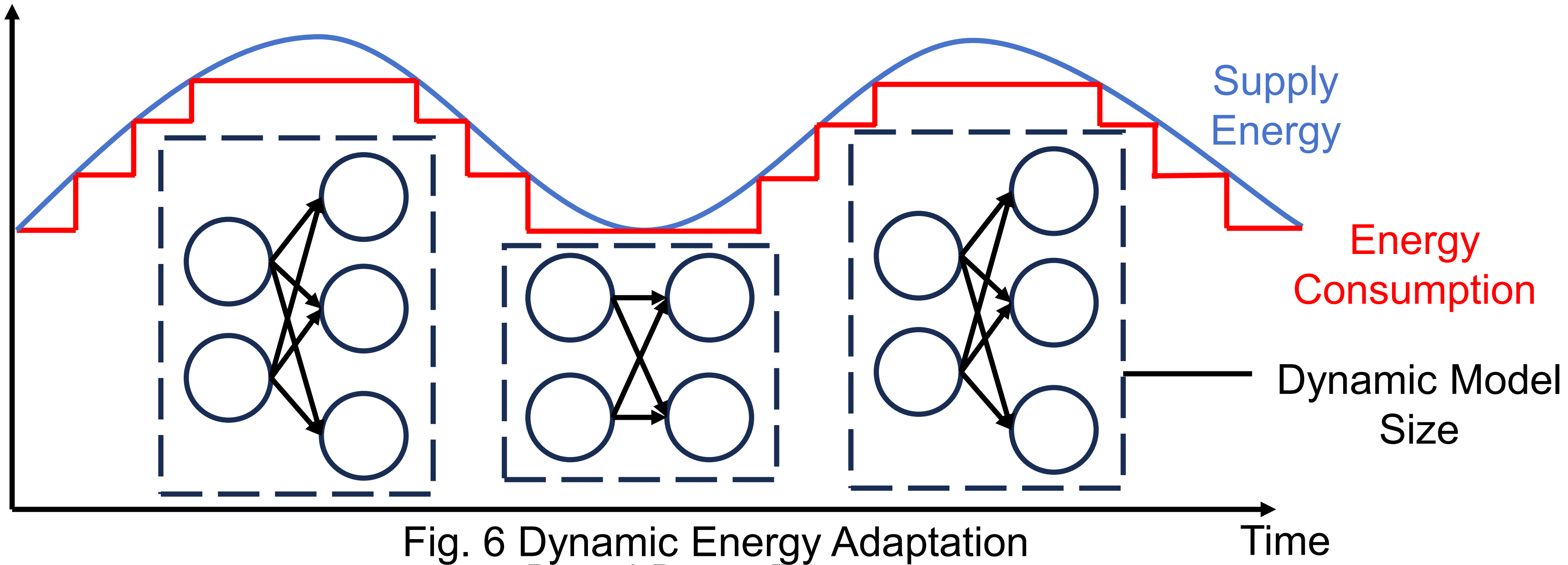


Fig. 6 Dynamic Energy Adaptation

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Overview

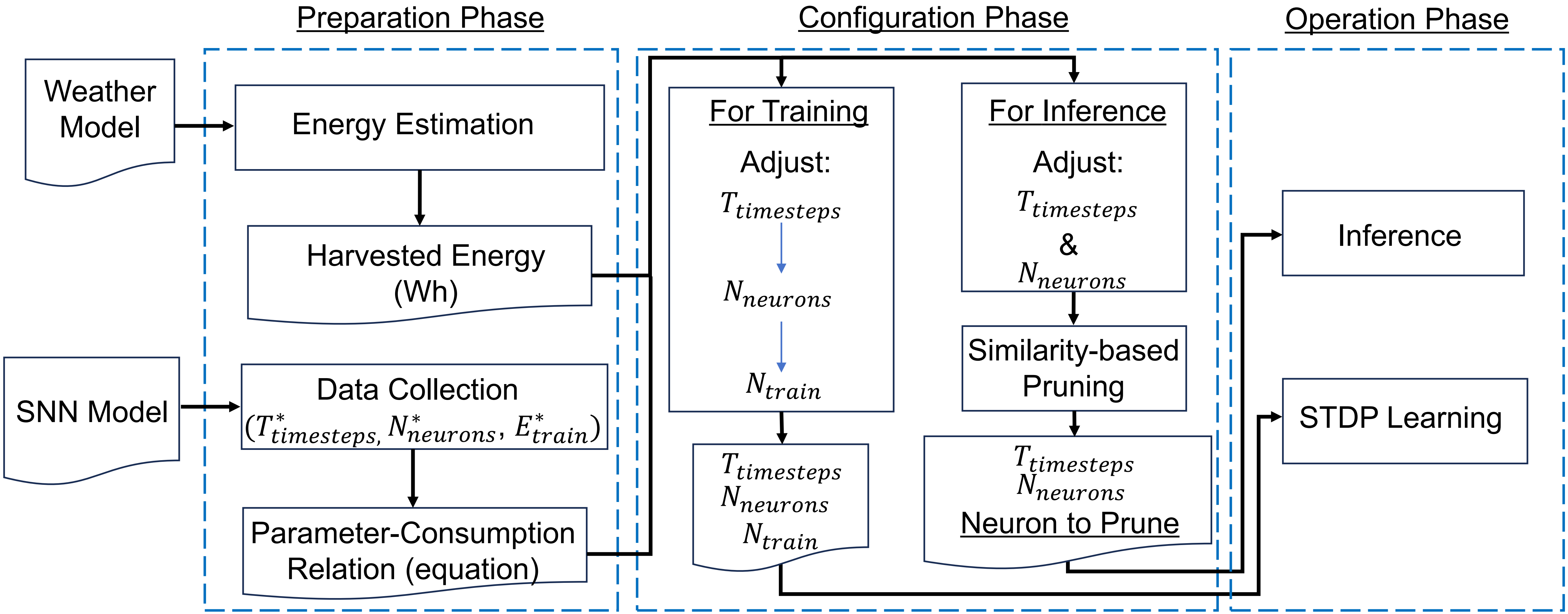


Fig. 7 Framework Overview

Overview

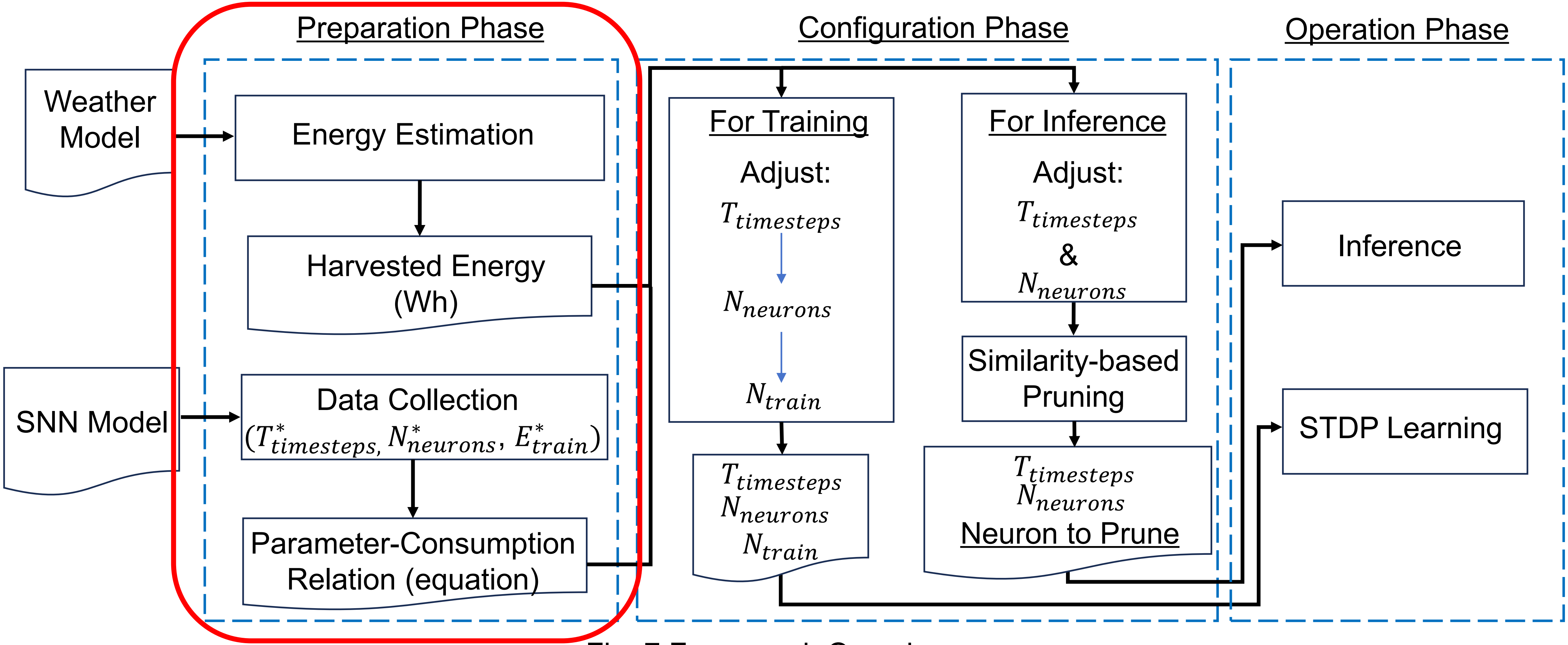


Fig. 7 Framework Overview

Preparation Phase (Energy Estimation)

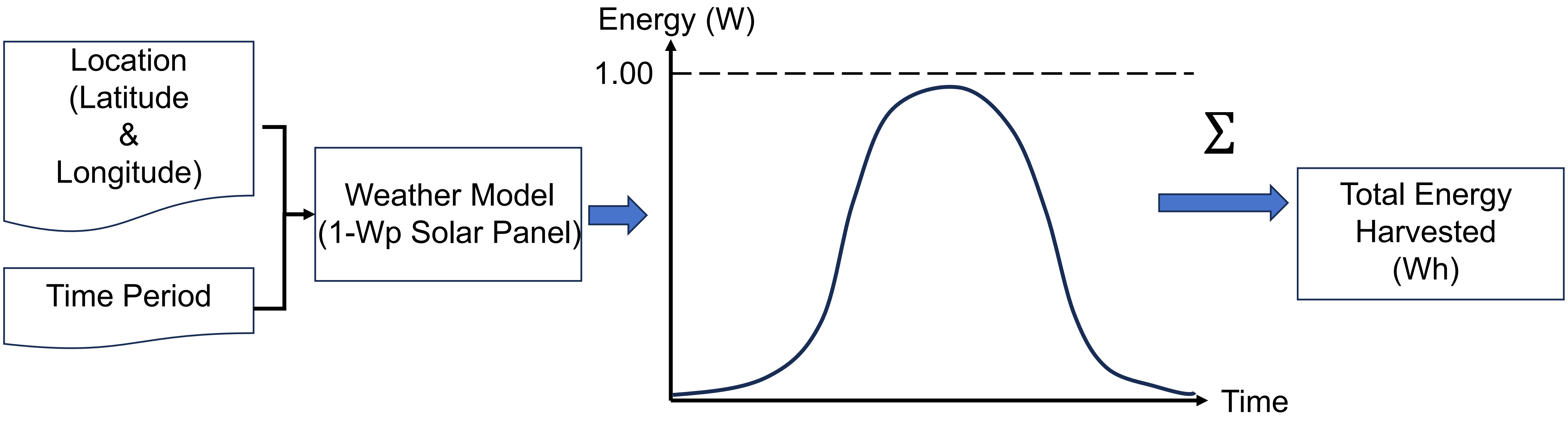


Fig. 8 Estimation of Harvested Energy

Preparation Phase (For Training)

Collect data points

$(T^*_{timesteps,} N_{neurons}$(fixed), $E^*_{train})$ → Derive general equation

$$T_{timesteps} = a \times E_{train} + b$$

Linear relation [9]

$(T_{timesteps}$(fixed), $N^*_{neurons}, \mathrm{E}^*_{\mathrm{train}})$ →

$$N_{neurons} = a \times E^d_{train} + b \times E_{train} + c$$

Non-linear relation [9]

Preparation Phase (For Inference)

Collect data points

$$(T^*_{timesteps,} N^*_{neurons}, E^*_{train})$$

Derive general equation

$$\mathrm{E}(\mathrm{N}_{\mathrm{neurons}}, \mathrm{T}_{\mathrm{timesteps}}) = aN^d_{neurons} + bT_{timesteps} + C$$

Overview

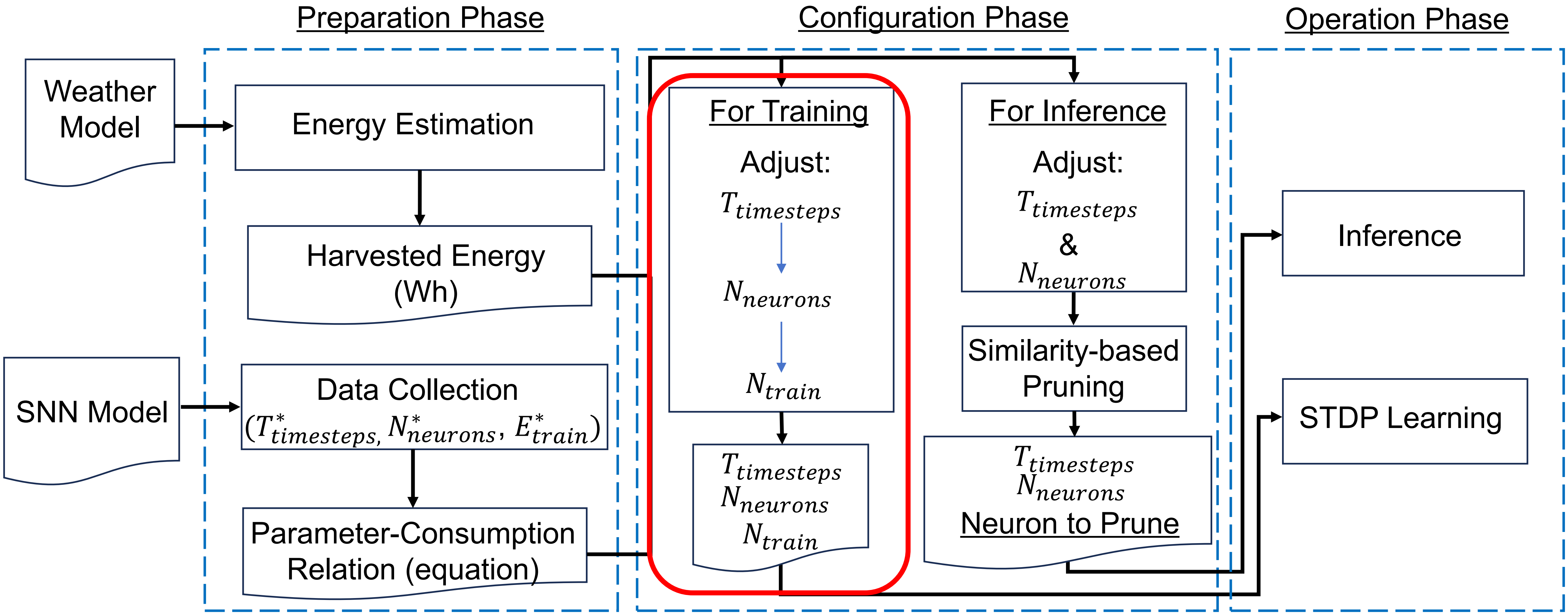


Fig. 7 Framework Overview

Configuration Phase

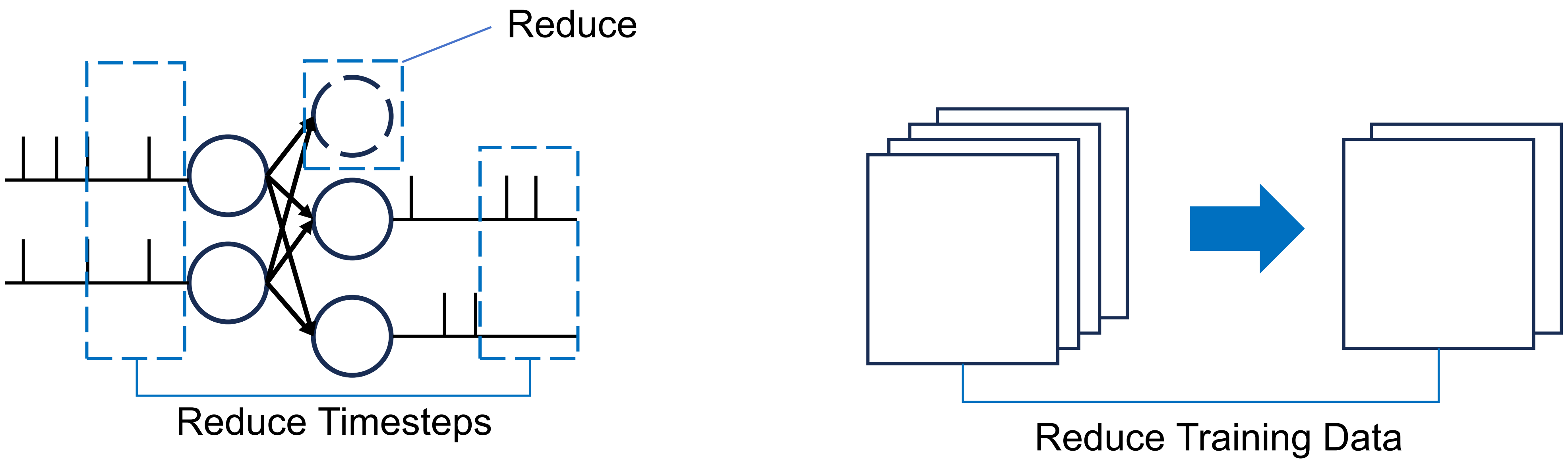


Fig. 12 Timestep Reduction and Pruning

Configuration Phase (Training)

Adjust number of timesteps

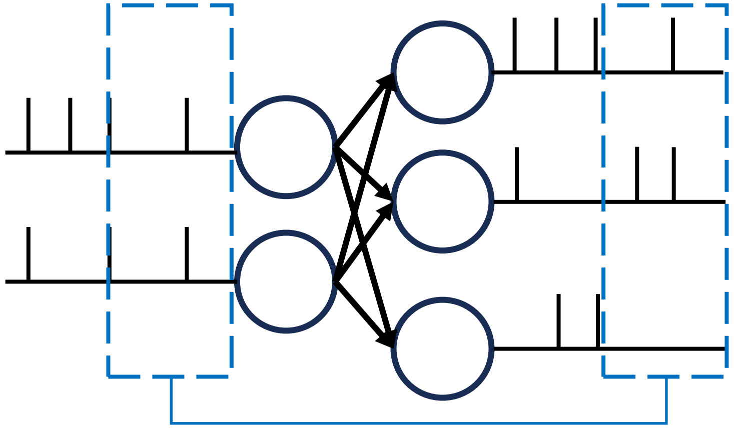
$$T_{timesteps} = a \times \boxed{E_{train}} + b$$

Total Energy Harvested

Adjust number of neurons

$$N_{neurons} = a \times \boxed{E_{train}^{d}} + b \times \boxed{E_{train}} + c$$

Total Energy Harvested

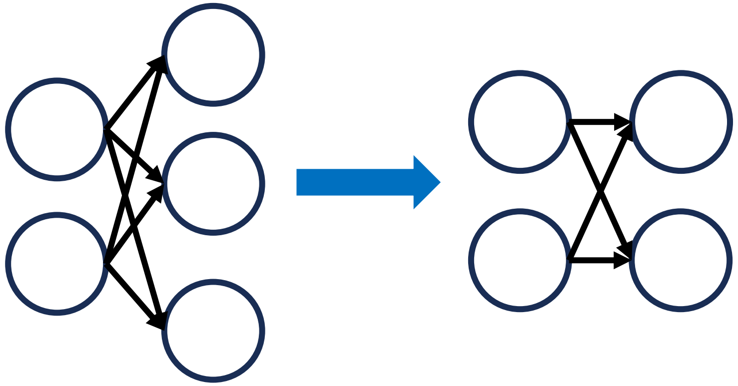


Reduce Timesteps

If

$T_{timesteps} < T_{minimum}$

$T_{timesteps} = T_{minimum}$



Reduce Number of Neuron

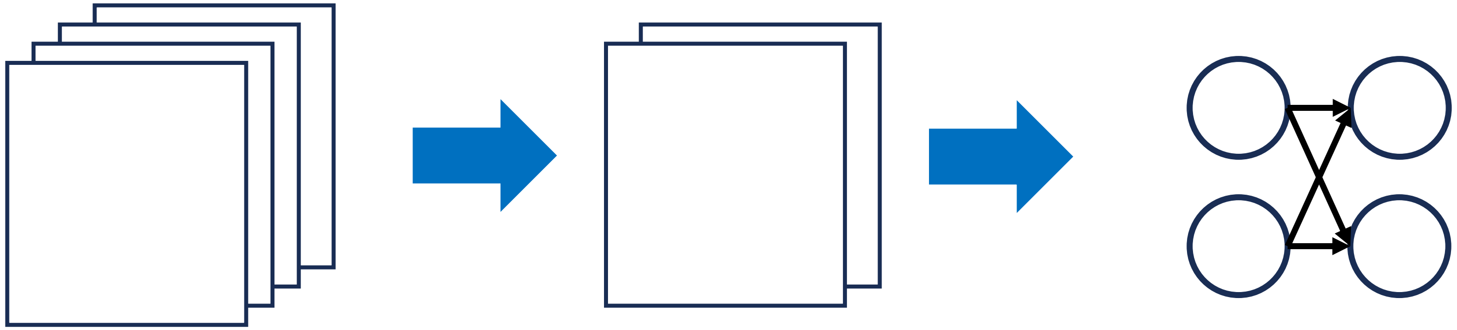
Fig. 9 Timesteps and Neurons Configuration

Configuration Phase (Training)(cnt.)

If

$N_{neurons} < N_{minimum}$

Reduce Number of
Training Data N_{train}



1. $N_{train} = N_{train} \times \frac{N_{neurons}}{N_{minimum}}$
2. $N_{neurons} = N_{minimum}$

Fig. 10 Training Data Reduction

Overview

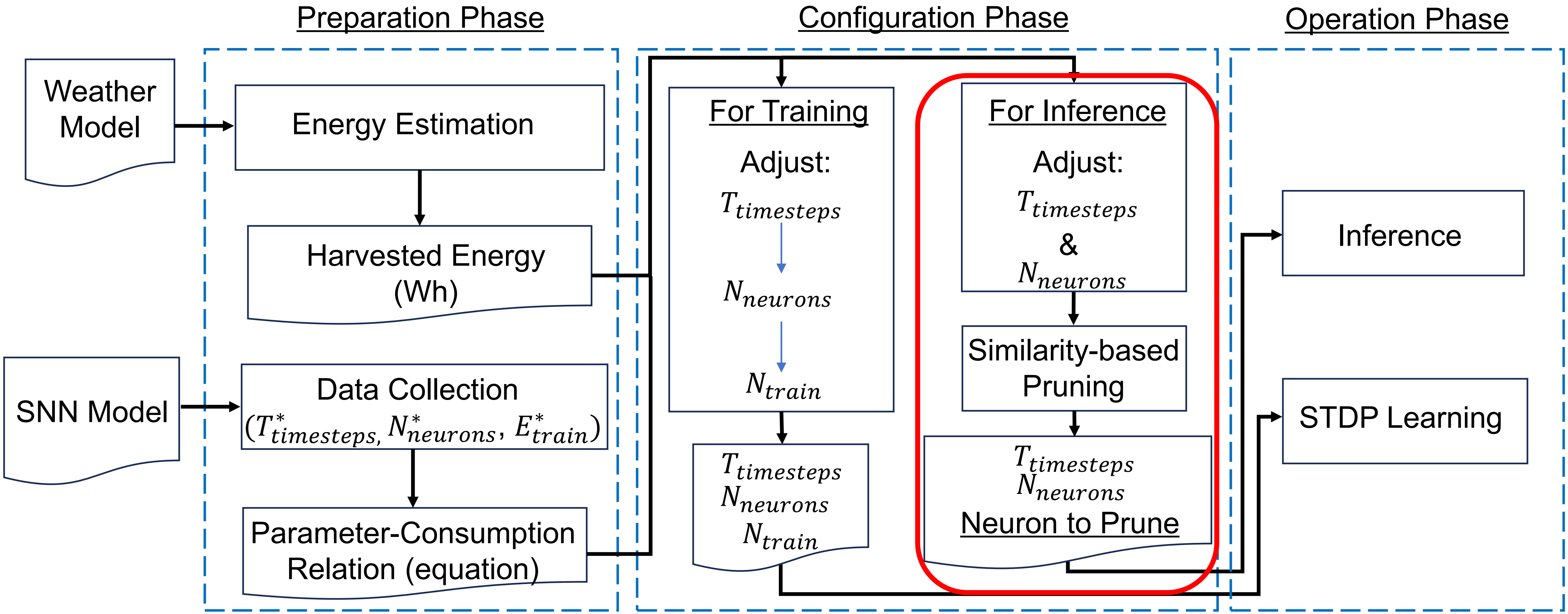


Fig. 7 Framework Overview

Configuration Phase (Inference)

$$\mathrm{E}(\mathrm{N}_{\mathrm{neurons}}, \mathrm{T}_{\mathrm{timesteps}}) = aN_{neurons}^{d} + bT_{timesteps} + C$$

Sequential-Least-Squares-Programming

$$\min_{N_{neurons}, T_{timesteps}} \left[\boxed{E_{inference}} - E\left(N_{neurons}, T_{timesteps}\right)\right]^2$$

Total Energy Harvested

$$T_{timesteps}, N_{neurons}$$

Configuration Phase (Inference)(cnt.)

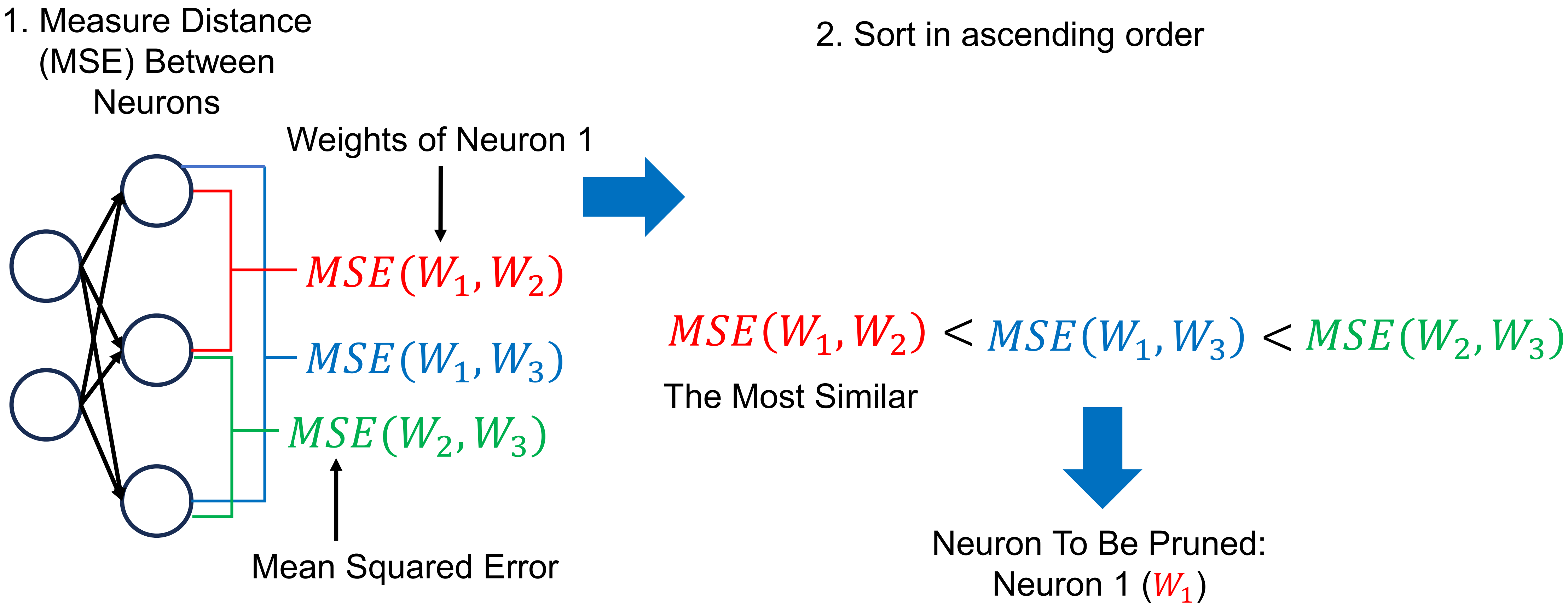


Fig. 11 Similarity-Based Neuron Pruning

Configuration Phase (Inference)(cnt.)

Adjust Timesteps & Prune Neuron

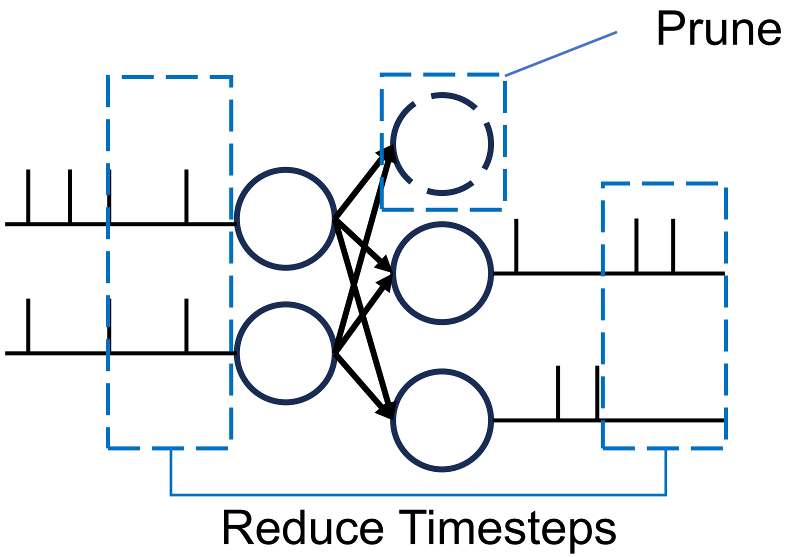


Fig. 12 Timestep Reduction and Pruning

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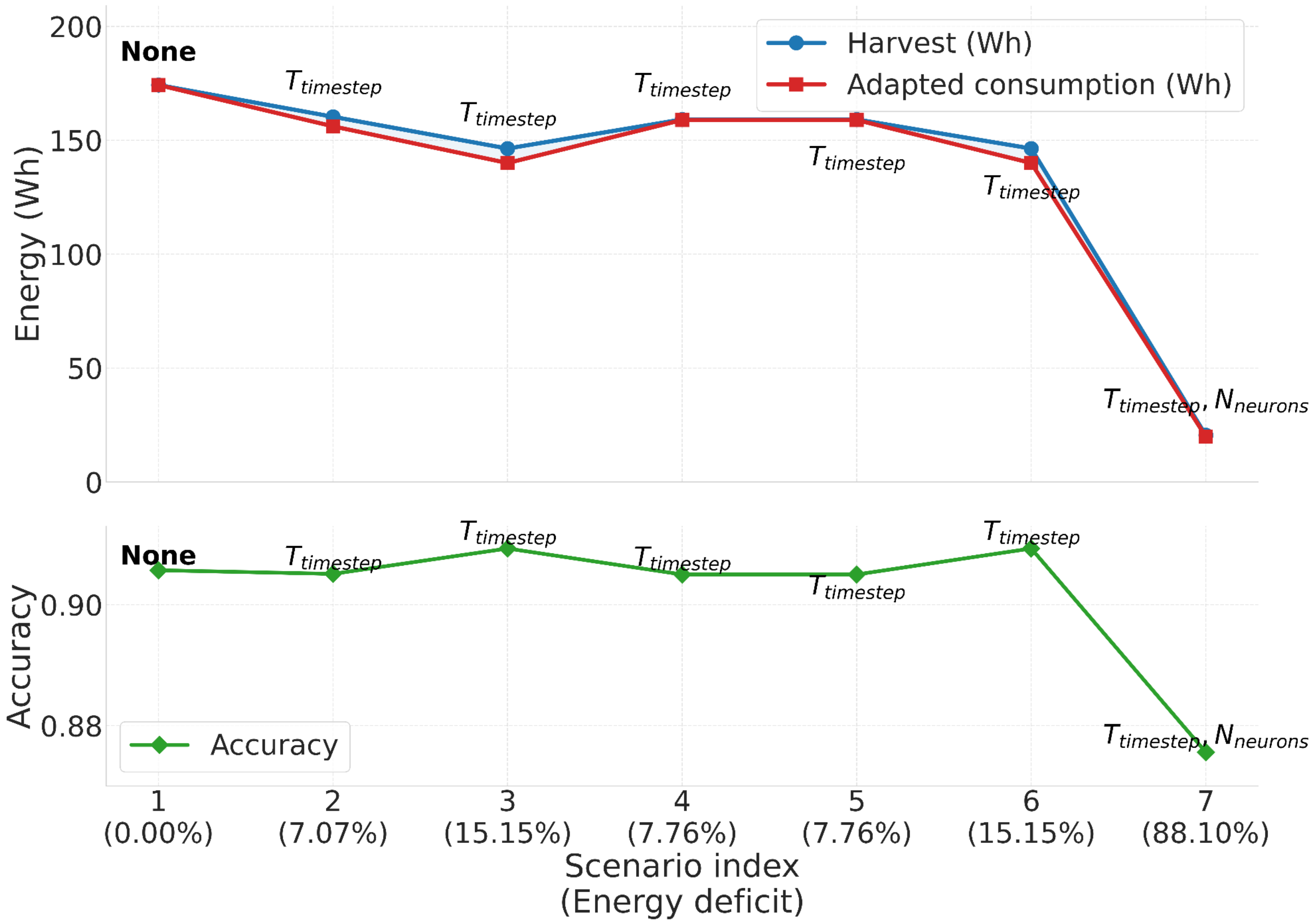
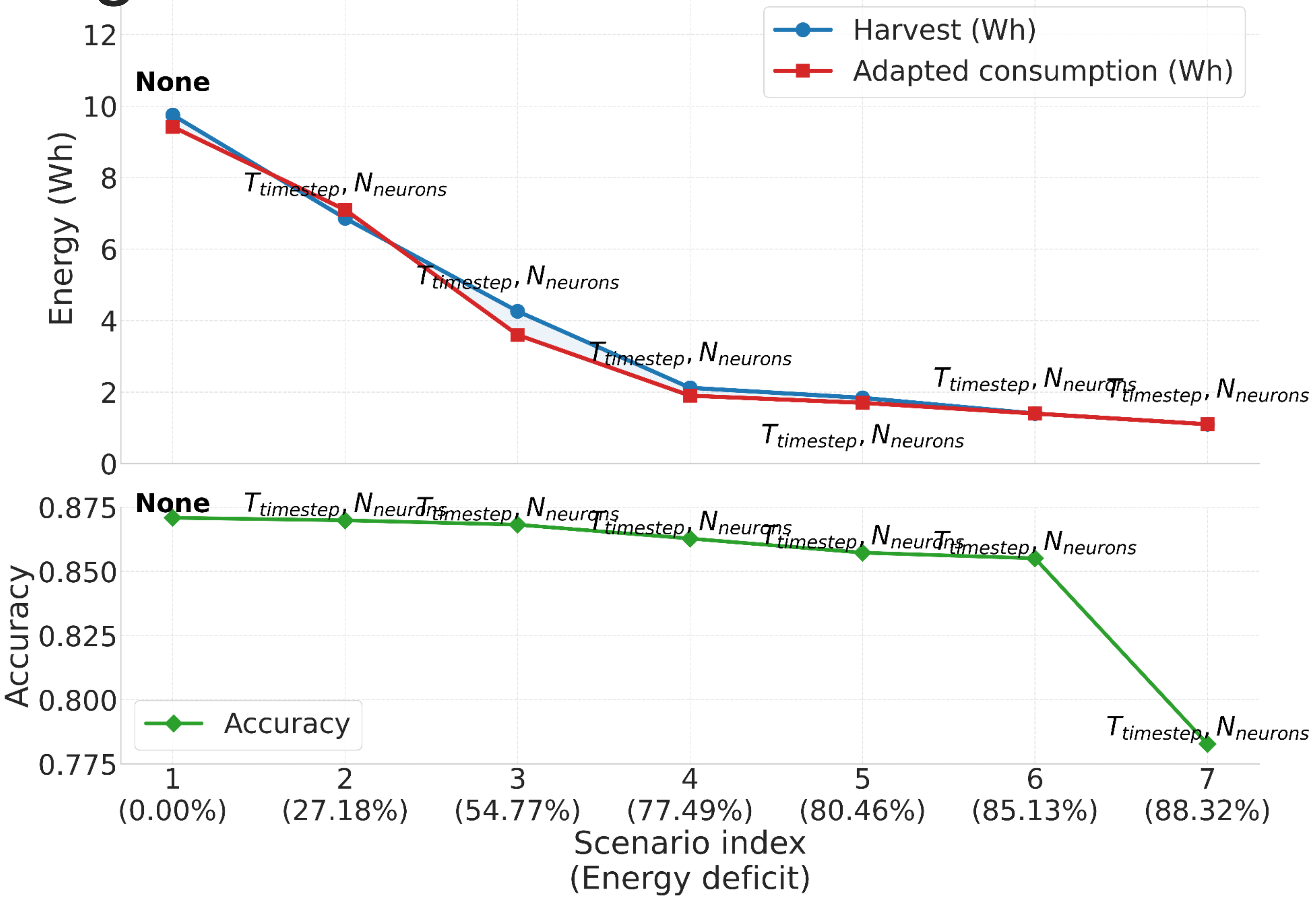
Evaluation Setup

- Implemented in: Python and BindsNet [10] Library
- Device: Raspberry Pi 5
- Model Architecture: Single Layer Feedforward with Lateral Inhibitory Signal
- Training Method: Spike-Timing-Dependent-Plasticity [8]
- Weather Condition: Open-Sourced Quartz-Solar-Forecast [11]
- Battery Simulation: Battery-Dataset-Preprocessing-Code-Library [12]
- Evaluation: Single model & Ensemble Learning [13]

Single 700-neuron Model Training



Single 700-neuron Model Inference



Ensemble Learning

Evaluation Setup: 70-neuron * 10 nodes

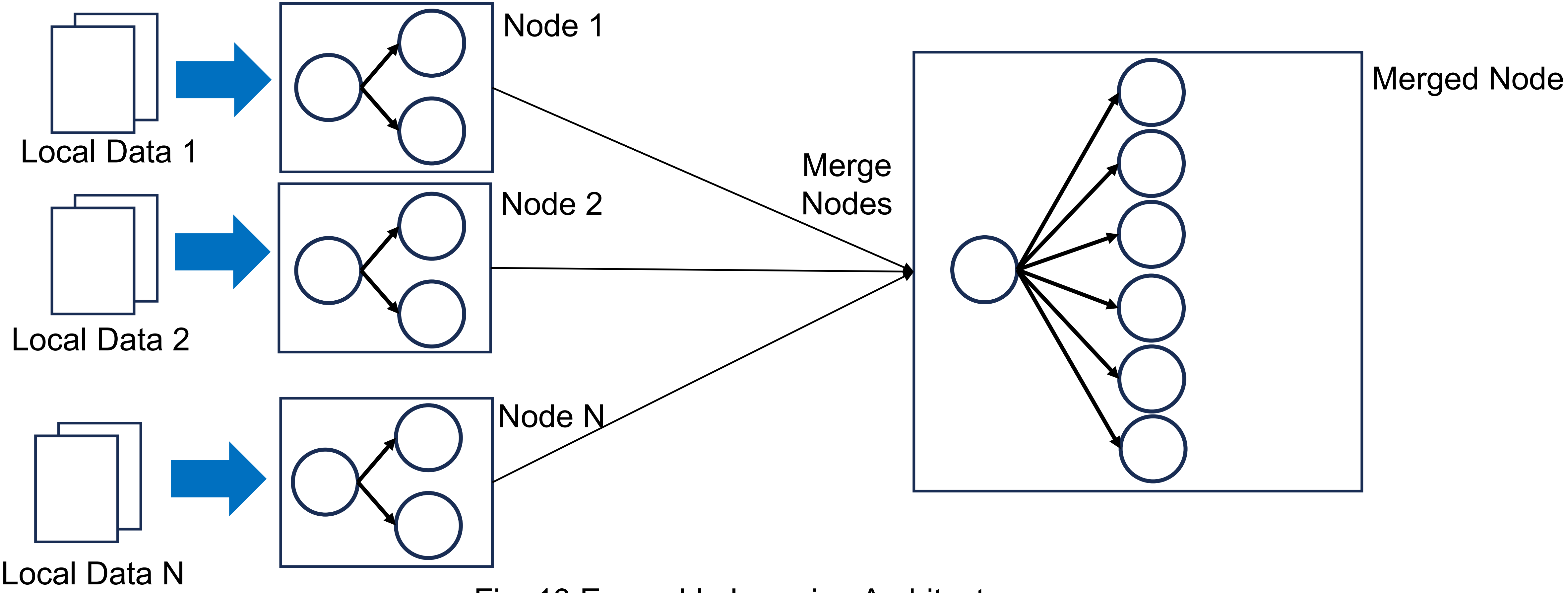
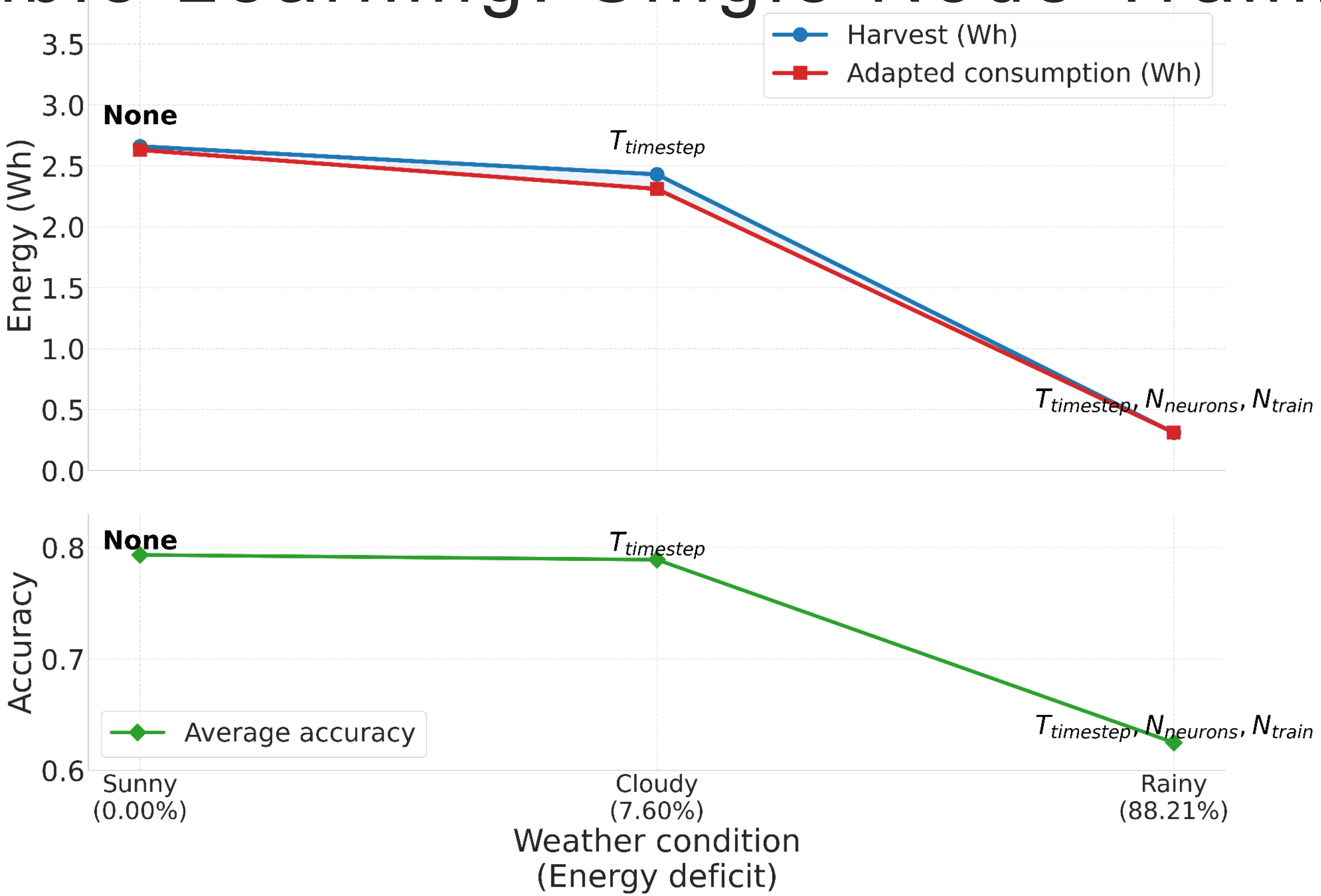
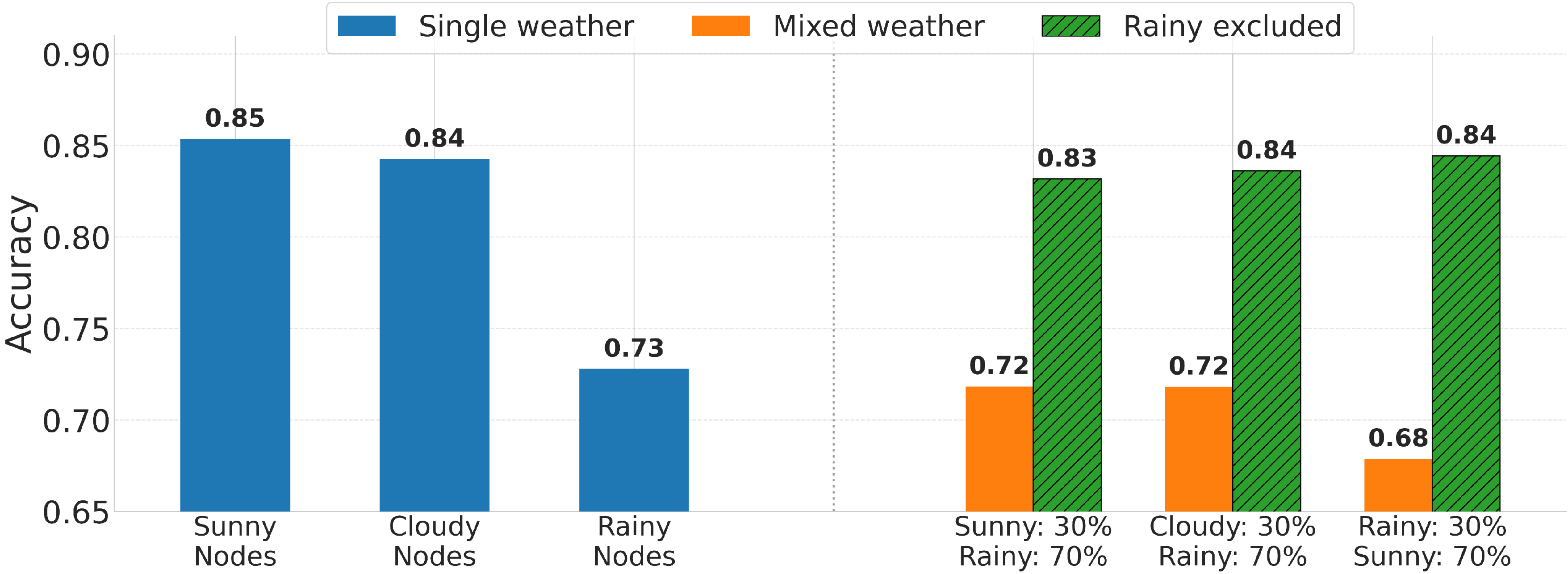


Fig. 13 Ensemble Learning Architecture

Ensemble Learning: Single Node Training



Ensemble Learning: Merged Node



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Conclusion

- The framework offers best AI performance under fluctuating energy supply.

- The framework dynamically reduces SNN energy consumption up to 85.13% with 1.5% accuracy loss.

- The framework will be implemented on FPGA for demonstration.

Schedule

	Aug.	Sep.	Oct.	Nov.	Dec.	Jan.	Feb.
FPGA SNN Implementation							
Dynamic Pruning by Clock Gating							
Performance Verification (SW & HW)							
FPGA Demonstration under Solar Energy							
Thesis Writing							

Published Papers

This work is based on the following papers:

- Yuga Hanyu, Subbaiah Ravi Hariprakash, Abderazek Ben Abdallah, Zhishang Wang, Khanh N. Dang, “GreenMorph: Sustainable Neuromorphic Computing Through Energy-Harvesting and Energy-Driven Online STDP Learning,” 2026 IEEE International Symposium on Circuits and Systems (ISCAS 2026)
- Yuga Hanyu, Khanh N. Dang, “EnsembleSTDP: Distributed In-situ Spike Timing Dependent Plasticity Learning in Spiking Neural Networks,” 2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC)
- Yuga Hanyu, Zhishang Wang, Khanh N. Dang, “Evolutionary Algorithm for STDP-Based Spiking Neural Network Model Compression,” 2025 IEEE 18th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC)

Reference

[1] "Edge AI revenue will reach USD157bn by 2030 globally, STL Partners predicts in brand new forecast," STL Partners, Jan. 29, 2025. [Online]. Available: https://stlpartners.com/press/edge-ai-revenue-to-reach-usd157bn-2030/

[2] A. Bamisile et al., "The environmental factors affecting solar photovoltaic output," Renewable and Sustainable Energy Reviews, vol. 208, Art. no. 115073, Jan. 2025, doi: 10.1016/j.rser.2024.115073. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1364032124007998

[3] M. Ligorati, J. Geyer-Klingeberg, T. E. Günther, and A. W. Rathgeber, "Degradation of lithium-ion batteries: A meta-analysis," Chemical Engineering Journal, vol. 506, Art. no. 158947, 2026, doi: 10.1016/j.cej.2026.158947. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S138589472601987X.

Reference

[4] D. Blalock et al., “What is the state of neural network pruning?” Proceedings of machine learning and systems, vol. 2, pp. 129–146, 2020

[5] T. Kolpe et al., “Enabling Improved Power Management in Multicore Processors through Clustered DVFS,” in 2011 Design, Automation & Test in Europe. IEEE, 2011, pp. 1–6

[6] M. Nagel et al., “A white paper on neural network quantization,” arXiv preprint arXiv:2106.08295, 2021

[7] A. B. Abdallah and K. N. Dang, “Neuromorphic computing principles and organization,” 2022

[8] P. U. Diehl and M. Cook, “Unsupervised learning of digit recognition using spike-timing-dependent plasticity,” Frontiers in computational neuroscience, vol. 9, p. 99, 201

Reference

[9] Z. Yan, Z. Bai, and W.-F. Wong, “Reconsidering the energy efficiency of spiking neural networks,” arXiv preprint arXiv:2409.08290, 2024

[10] H. Hazan et al., “Bindsnet: A machine learning-oriented spiking neural networks library in python,” Frontiers in neuroinformatics, vol. 12, p. 89, 2018

[11] P. Dudfield et al., “open-source-quartz-solar-forecast,” Mar. 2025. [Online]. Available: https://github.com/openclimatefix/open-source-quartz-solar-forecast

[12] F. Wang et al., “Physics-Informed Neural Network for Lithium-Ion Battery Degradation Stable Modeling and Prognosis,” Nature Communications, vol. 15, no. 1, p. 4332, 20

[13] H. Yuga and K. N. Dang, “EnsembleSTDP: Distributed in-situ spike timing dependent plasticity learning in spiking neural networks,” in 2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC). IEEE, 2024, pp. 434–440

Thank you for your attention!